

By Theorem 10 from Chapter 1, A^{-1} is the standard matrix of T^{-1} .

From Theorem 6(b), A^{-1} is invertible.

By the Invertible Matrix Theorem and the equivalence of the statements (a), (f) and (i), T^{-1} is one-to-one and also maps $\mathbb{R}^n$ onto $\mathbb{R}^n$. II

37) Suppose T and U are linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^n$ such that $T(U(x)) = x$ for all x in $\mathbb{R}^n$. Is it true that $U(T(x)) = x$ for all x in $\mathbb{R}^n$? Why or why not?

Solution: By Theorem 10, from Chapter 1, A and B are standard $n \times n$ matrices of T and U respectively such that $T(x) = Ax$ and $U(x) = Bx$ for all $x \in \mathbb{R}^n$.
Given $T(U(x)) = x$ for all x in $\mathbb{R}^n$
 $\therefore T(Bx) = x \Rightarrow A(Bx) = x \Rightarrow (AB)x = x$ (Theorem 2(a))

So, for all $x \in \mathbb{R}^n$, there exists at least one vector x in $\mathbb{R}^n$ s.t. $(AB)x = x$.
 $\therefore$ By Defn. 2.0 $\Rightarrow (AB)x$ is onto $\mathbb{R}^n$ to $\mathbb{R}^n$.

By the Invertible Matrix Theorem and the equivalence of the statements (i) and (a), we know that AB is invertible.

By Problems 21 and 28 we know that both A and B are invertible.
From Theorem 6(b), BA is invertible and $(BA)^{-1} = A^{-1}B^{-1}$

From the defn. of identity matrix, $AB = I_n$.
By Problem 25, A is invertible and B is invertible with $B = A^{-1}$ and $A = B^{-1}$.
 $\therefore BA = BB^{-1} = I_n$ (By the defn. of inverse)

$(BA)x = I_n x \Rightarrow (BA)x = x$ (By the defn. of identity matrix)
 $\Rightarrow B(Ax) = x$ (Theorem 2(a)) $\Rightarrow U(Ax) = x \Rightarrow \boxed{U(T(x)) = x} \square$

The last 2 steps follow from Theorem 10

from Chapter 1. ~~II~~

38) Suppose a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ has the property that $T(u) = T(v)$ for some pair of distinct vectors u and v in $\mathbb{R}^n$. Can T map $\mathbb{R}^n$ onto $\mathbb{R}^n$? Why or why not?